ITA – 0510 COMPUTER VISION ASSIGNMENT CODE

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```html
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport"
     content="width=device-width, initial-scale=1.0">

  <title>Fruit Quality Grading</title>

  <style>

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
      font-family: Arial, Helvetica, sans-serif;
    }

    body {
      background: #f4f7f5;
      color: #222;
```

```css
}

header {
  background: linear-gradient(
    135deg,
    #1b5e20,
    #43a047
  );
  color: white;
  padding: 25px;
  text-align: center;
}

header h1 {
  font-size: 32px;
  margin-bottom: 8px;
}

header p {
  font-size: 16px;
}

.container {
  width: 92%;
  max-width: 1200px;
  margin: 30px auto;
}
```

```css
.card {
  background: white;
  padding: 25px;
  margin-bottom: 25px;
  border-radius: 15px;
  box-shadow:
    0 4px 15px rgba(0,0,0,0.08);
}

.card h2 {
  color: #2e7d32;
  margin-bottom: 20px;
}

.upload-area {
  border: 3px dashed #66bb6a;
  padding: 35px;
  text-align: center;
  border-radius: 15px;
  background: #f1f8e9;
}

input[type="file"] {
  margin: 20px 0;
  padding: 10px;
}

button {
```

```css
    background: #2e7d32;
    color: white;
    border: none;
    padding: 12px 22px;
    border-radius: 8px;
    cursor: pointer;
    font-size: 15px;
    margin: 5px;
}

button:hover {
    background: #1b5e20;
}

.danger {
    background: #c62828;
}

.danger:hover {
    background: #8e0000;
}

.image-container {
    display: grid;
    grid-template-columns:
        repeat(auto-fit, minmax(250px, 1fr));
    gap: 20px;
    margin-top: 20px;
```

```css
}

.image-box {
   text-align: center;
   background: #fafafa;
   padding: 15px;
   border-radius: 12px;
}

.image-box h3 {
   margin-bottom: 10px;
   color: #444;
}

.image-box img,
.image-box canvas {
   width: 100%;
   max-height: 300px;
   object-fit: contain;
   border-radius: 10px;
}

.metrics {
   display: grid;
   grid-template-columns:
      repeat(auto-fit, minmax(180px, 1fr));
   gap: 15px;
}
```

```css
.metric {
    background: #f1f8e9;
    padding: 20px;
    border-radius: 12px;
    text-align: center;
}

.metric h3 {
    color: #555;
    margin-bottom: 8px;
}

.metric span {
    font-size: 24px;
    font-weight: bold;
    color: #2e7d32;
}

.result {
    text-align: center;
    padding: 30px;
    border-radius: 15px;
    margin-top: 20px;
}

.result.good {
    background: #e8f5e9;
```

```
    border: 2px solid #43a047;
}

.result.medium {
    background: #fff8e1;
    border: 2px solid #ffb300;
}

.result.bad {
    background: #ffebee;
    border: 2px solid #e53935;
}

.grade {
    font-size: 45px;
    font-weight: bold;
    margin: 15px;
}

.confidence {
    font-size: 20px;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 15px;
}
```

```css
th,
td {
  border: 1px solid #ddd;
  padding: 12px;
  text-align: center;
}

th {
  background: #2e7d32;
  color: white;
}

tr:nth-child(even) {
  background: #f5f5f5;
}

.progress-container {
  width: 100%;
  background: #ddd;
  border-radius: 10px;
  overflow: hidden;
  margin-top: 10px;
}

.progress {
  height: 20px;
  width: 0%;
```

```
  background: #43a047;
  transition: width 1s;
}

.hidden {
  display: none;
}

.status {
  text-align: center;
  padding: 15px;
  color: #555;
}

footer {
  background: #1b5e20;
  color: white;
  text-align: center;
  padding: 20px;
  margin-top: 40px;
}

@media(max-width: 600px) {

  header h1 {
    font-size: 24px;
  }
```

```
        .container {
          width: 95%;
        }

        .card {
          padding: 18px;
        }

      }

    </style>

  </head>



  <body>



  <header>

    <h1>
      🍎 FOOD PROCESSING – FRUIT QUALITY GRADING
    </h1>

    <p>
      Computer Vision Based Fruit Quality Inspection System
    </p>
```

```
</header>


<div class="container">


  <!-- =====================================================
     UPLOAD
  ====================================================== -->

  <div class="card">

    <h2>
      1. Upload Fruit Image
    </h2>

    <div class="upload-area">

      <p>
        Upload an image of a fruit for quality analysis.
      </p>

      <input
        type="file"
        id="imageInput"
        accept="image/*"
      >
```

```html
    <br>

    <button onclick="analyzeFruit()">
      🔍 Analyze Fruit
    </button>

    <button
      class="danger"
      onclick="resetSystem()"
    >
      Reset
    </button>

  </div>

</div>


<!-- ======================================================
   IMAGE PREVIEW
======================================================= -->

<div
  class="card hidden"
  id="imageSection"
>

  <h2>
```

```
  2. Image Preprocessing
</h2>

<div class="image-container">

  <div class="image-box">

    <h3>
      Original Image
    </h3>

    <img id="originalImage">

  </div>



  <div class="image-box">

    <h3>
      Resized Image
    </h3>

    <canvas id="resizeCanvas"></canvas>

  </div>



  <div class="image-box">
```

```
        <h3>
          Enhanced Image
        </h3>

        <canvas id="enhancedCanvas"></canvas>

      </div>


      <div class="image-box">

        <h3>
          Segmented Fruit
        </h3>

        <canvas id="segmentCanvas"></canvas>

      </div>

    </div>

</div>


<!-- ========================================================
    FEATURE ANALYSIS
======================================================== -->
```

```
<div
  class="card hidden"
  id="featureSection"
>

  <h2>
    3. Feature Analysis
  </h2>

  <div class="metrics">

    <div class="metric">

      <h3>
        Mean Hue
      </h3>

      <span id="hueValue">
        -
      </span>

    </div>


    <div class="metric">

      <h3>
```

```
    Saturation
  </h3>

  <span id="saturationValue">
    -
  </span>

</div>


<div class="metric">

  <h3>
    Brightness
  </h3>

  <span id="brightnessValue">
    -
  </span>

</div>


<div class="metric">

  <h3>
    Fruit Area
  </h3>
```

```
  <span id="areaValue">
    -
  </span>

</div>


<div class="metric">

  <h3>
    Width
  </h3>

  <span id="widthValue">
    -
  </span>

</div>


<div class="metric">

  <h3>
    Height
  </h3>

  <span id="heightValue">
```

```
    -
  </span>

</div>


<div class="metric">

  <h3>
    Circularity
  </h3>

  <span id="circularityValue">
    -
  </span>

</div>


<div class="metric">

  <h3>
    Surface Damage
  </h3>

  <span id="damageValue">
    -
  </span>
```

```
    </div>

  </div>

</div>



<!-- ====================================================
   TRADITIONAL METHOD
===================================================== -->

<div
  class="card hidden"
  id="traditionalSection"
>

  <h2>
    4. Traditional Computer Vision Grading
  </h2>

  <p>
    The traditional approach uses colour, shape,
    size, brightness and surface-damage measurements.
  </p>

  <div
    class="result"
```

```
      id="traditionalResult"
    >

      <div>
        Traditional Quality Grade
      </div>

      <div
        class="grade"
        id="traditionalGrade"
      >
        -
      </div>

    </div>

  </div>



  <!-- ====================================================
      DEEP LEARNING
  ===================================================== -->

  <div
    class="card hidden"
    id="deepLearningSection"
  >
```

```html
<h2>
  5. Deep Learning Classification
</h2>

<p>
  MobileNetV2 transfer-learning architecture is
  proposed for automated fruit quality classification.
</p>

<br>

<p>
  Model Processing:
</p>

<div class="progress-container">

  <div
    class="progress"
    id="progressBar"
  ></div>

</div>

<div
  class="status"
  id="processingStatus"
>
```

```html
    Processing image...
  </div>

</div>



<!-- ====================================================
   FINAL RESULT
===================================================== -->

<div
  class="card hidden"
  id="resultSection"
>

  <h2>
    6. Final Quality Result
  </h2>

  <div
    class="result"
    id="finalResult"
  >

    <p>
      Predicted Fruit Quality
    </p>
```

```
      <div
        class="grade"
        id="finalGrade"
      >
        -
      </div>

      <div class="confidence">

        Confidence:
        <strong id="confidenceValue">
          -
        </strong>

      </div>

    </div>

  </div>



  <!-- ========================================================
      COMPARISON
  ======================================================== -->

  <div
    class="card hidden"
    id="comparisonSection"
```

>

<h2>

7. Traditional vs Deep Learning

</h2>

<table>

<thead>

<tr>

<th>

Parameter

</th>

<th>

Traditional Method

</th>

<th>

Deep Learning

</th>

</tr>

</thead>

<tbody>

<tr>

<td>

Accuracy

</td>

<td>

Depends on thresholds

</td>

<td>

Learns from training data

</td>

</tr>

<tr>

<td>

Robustness

</td>

<td>

Lower

</td>

<td>

Higher

</td>

</tr>

<tr>

<td>

Training Requirement

</td>

<td>

Low

</td>

<td>

High

</td>

</tr>

<tr>

<td>

Computational Cost

</td>

<td>

Low

</td>

<td>

Medium / High

</td>

</tr>

<tr>

<td>

Natural Variations

</td>

<td>

Difficult

</td>

<td>

Better Adaptation

</td>

</tr>

<tr>

<td>

Production Suitability

</td>

<td>

Controlled conditions

</td>

<td>

Variable conditions

</td>

</tr>

</tbody>

</table>

</div>

<!-- ==

```
 QUALITY REPORT
========================================================== -->

<div
  class="card hidden"
  id="reportSection"
>

  <h2>
    8. Quality Inspection Report
  </h2>

  <table>

    <tbody>

      <tr>

        <th>
          Parameter
        </th>

        <th>
          Result
        </th>

      </tr>
```

```
<tr>

  <td>
    Fruit Quality
  </td>

  <td id="reportGrade">
    -
  </td>

</tr>


<tr>

  <td>
    Confidence
  </td>

  <td id="reportConfidence">
    -
  </td>

</tr>


<tr>
```

```
  <td>
    Surface Damage
  </td>

  <td id="reportDamage">
    -
  </td>

</tr>


<tr>

  <td>
    Fruit Area
  </td>

  <td id="reportArea">
    -
  </td>

</tr>


<tr>

  <td>
```

```
            Circularity
          </td>

          <td id="reportCircularity">
            -
          </td>

        </tr>


        <tr>

          <td>
            Inspection Status
          </td>

          <td id="inspectionStatus">
            -
          </td>

        </tr>

      </tbody>

    </table>


    <br>
```

```
    <button onclick="downloadReport()">
      📄 Download Report
    </button>

  </div>

</div>

<footer>

  <p>
    Food Processing – Fruit Quality Grading
  </p>

  <p>
    Computer Vision Project | ITA05
  </p>

</footer>

<script>

// ============================================================
```

```
// GLOBAL VARIABLES
// ============================================================

let currentData = {};



// ============================================================
// IMAGE ANALYSIS
// ============================================================

function analyzeFruit() {

   const input =
      document.getElementById(
         "imageInput"
      );

   if (
      !input.files ||
      input.files.length === 0
   ) {

      alert(
         "Please upload a fruit image first."
      );

      return;
   }
```

```
};

const file = input.files[0];

const reader =
  new FileReader();


reader.onload = function(event) {

  const image =
    document.getElementById(
      "originalImage"
    );

  image.src =
    event.target.result;


  image.onload = function() {

    processImage(image);

  };

};
```

```
  reader.readAsDataURL(file);

}


// ==========================================================
// PROCESS IMAGE
// ==========================================================

function processImage(image) {

  document
    .getElementById("imageSection")
    .classList.remove("hidden");

  document
    .getElementById("featureSection")
    .classList.remove("hidden");

  document
    .getElementById("traditionalSection")
    .classList.remove("hidden");

  document
    .getElementById("deepLearningSection")
    .classList.remove("hidden");

  document
```

```
  .getElementById("resultSection")
  .classList.remove("hidden");

document
  .getElementById("comparisonSection")
  .classList.remove("hidden");

document
  .getElementById("reportSection")
  .classList.remove("hidden");



// --------------------------------------------------------
// Resize
// --------------------------------------------------------

const resizeCanvas =
  document.getElementById(
    "resizeCanvas"
  );

const resizeContext =
  resizeCanvas.getContext(
    "2d"
  );



resizeCanvas.width = 300;
```

```
resizeCanvas.height = 300;


resizeContext.drawImage(
  image,
  0,
  0,
  300,
  300
);


// -------------------------------------------------------
// Enhanced image
// -------------------------------------------------------

const enhancedCanvas =
  document.getElementById(
    "enhancedCanvas"
  );

const enhancedContext =
  enhancedCanvas.getContext(
    "2d"
  );


enhancedCanvas.width = 300;
```

```
enhancedCanvas.height = 300;


enhancedContext.filter =
  "contrast(1.15) brightness(1.08) saturate(1.15)";

enhancedContext.drawImage(
  image,
  0,
  0,
  300,
  300
);


// ------------------------------------------------------
// Segmentation
// ------------------------------------------------------

const segmentCanvas =
  document.getElementById(
    "segmentCanvas"
  );

const segmentContext =
  segmentCanvas.getContext(
    "2d"
  );
```